Deakin DUPRF Fellowship — Planning Document

1. Project summary

An efficient, learning-based sensing framework for 5G/6G Integrated Sensing and Communication (ISAC) networks. Using signals the network already transmits, the framework detects, localises, and tracks objects. The two main contributions are:

1. **Receiver selection** — a learning-based method for choosing which receivers to activate for sensing.
2. **Multi-target tracking** — a robust method for maintaining target identities over time.

2. Objectives

- **O1.** Develop and validate the learning-based receiver-selection scheme.
- **O2.** Develop the hybrid multi-target tracking method.
- **O3.** Integrate and evaluate the full pipeline in a 3GPP-compliant setting.

3. Timeline (3 years)

Year 1 — Foundations

Period	**Task**
Q1	Set up simulation testbed; define metrics
Q2	Implement model-based selection evaluator (FIM / CRLB / GDOP)
Q3	Build tiered receiver-selection baseline
Q4	Two-pass pipeline working; first results

Milestone M1 (end of Year 1): working baseline, first results.

Year 2 — Core contributions

Period	**Task**
Q1	Train learning-based selection policy

Period	Task
Q2	Implement hybrid tracker
Q3	Ablation study: where learning helps
Q4	End-to-end integration; submit papers

Milestone M2 (end of Year 2): core methods complete and published.

Year 3 — Validation and thesis

Period	Task
Q1	Large-scale evaluation (density, height, trajectories)
Q2	Robustness and tuning
Q3	Final experiments and analysis
Q4	Thesis and final papers

Milestone M3 (end of Year 3): thesis and final results.

Note: the core contributions (O1 and O2) complete by the end of Year 2, so the third year is scaling and writing rather than new research.

4. Deliverables

- Working simulation framework (code in this repository)
- Conference and journal papers (target: 3-4)
- Thesis